

Profile

Machine learning engineer with a strong base in fine-tuning, VLMs, agentic systems, and applying ML in specialist domains. Outside my day-to-day work, I compete on Kaggle to explore unfamiliar problems, learn quickly from real constraints, and bring those lessons back to work when building models and tools.

Education

University of Cambridge, MPhil in Data Intensive Science (Distinction)	2024/10 - 2025/07
Advanced study in statistics, machine learning, high-performance computing, and data-intensive scientific applications.	
University of Edinburgh, BA in Mathematics (First Class Honours)	2020/09 - 2024/05
Mathematics degree with emphasis on statistics, machine learning, and mathematical foundations for modelling.	

Work Experience

Viridien Machine Learning Engineer	2025/08 – Present
VLM document classification: Fine-tuned vision-language models for oil & gas technical documents using LoRA SFT, RLVR methods (GRPO/GSPO), and prompt optimization with ACE-Agent and DSPy. Improved F1 / accuracy from 0.47 / 0.74 to 0.78 / 0.85 at comparable model size. Agentic Label QC System: Built an agentic QC system through close collaboration with domain experts. Translated expert feedback into agents improvements, incorporating vision capabilities, expert-written skills for domain knowledge injection, and a critic-validator framework to improve label reliability. Agent terminal-use training: Done Reinforcement Learning on Qwen models on multi-GPU infrastructure with Harbor, SGLang, and AREAL to improve terminal-use behavior, in collaboration with CAMEL-AI.	
Autodesk Intern, Technology Consultant (AI/ML)	2024/06 – 2024/09
VLM benchmarking for CAM: Created a benchmark dataset for evaluating how vision-language models understand Computer-Aided Manufacturing workflows. The benchmark supported model selection for Fusion agents and gives the team a repeatable way to test new model releases.	
Business-Intelligence of Oriental Nations Corporation Ltd Intern, Algorithm Engineer	2023/06 – 2023/09
Manufacturing ML: Trained machine learning models for manufacturing applications using scikit-learn and XGBoost.	

Selected Competitions & Hackathons

Kaggle Hackathon Honorable Mention (8/321) - [Solution Write-up]	May 2026
Fine-tuned Gemma 2 2B Instruct for reasoning with a curated data mixture, LoRA reasoning SFT, and DPO based on the Delta Learning Hypothesis. Trained on Google TPU.	
Kaggle Competition Master (Ranked 842 worldwide; 1x Gold, 2x Silver).	Feb 2026
Kaggle Competition Gold Medal (13/3357) - [Solution Write-up]	Feb 2026
Built a Rust solver for irregular 2D bin packing, placing non-convex shapes into the smallest possible bounding square across hundreds of configurations. Combined simulated annealing, local search, and constraint relaxation over the 3-month competition.	
Kaggle Competition Silver Medal (19/1151) - Presented at NeurIPS 2025 Special Session [Solution Presentation]	Oct 2025
Built a program-synthesis agent with an evolutionary database to solve ARC-AGI tasks while optimizing for minimal code length.	
Mistral AI Award - Hack the Law Hackathon	Jun 2025
Built a Mixtral-based multi-agent system to extract, classify, and link regulated activities and entities from legal documents.	
1 st Place - Infosys Cambridge AI Centre Hackathon	May 2025
Built Memory Plus , an MCP memory layer for LLMs with a local vector store and RAG. (50+ GitHub stars; 8k downloads)	

Publications

SeLoRA: Self-Expanding LoRA for High-Quality and Efficient Medical Image Synthesis	Feb 2026
Expert Systems with Applications • Project Page	
Semi-Supervised Medical Image Segmentation via Knowledge Mining from Large Models	Jan 2026
arXiv Preprint • Accepted to IEEE International Symposium on Biomedical Imaging (ISBI)	

Skills

AI/ML	Fine-tuning, LoRA, RLVR/GRPO, DPO, prompt optimization, agent evaluation, RAG, synthetic data.
Languages & Tools	Python, Git, Linux, Docker, Neo4j, Qdrant, SQL, LaTeX.
Libraries	vLLM, FastAPI, PyTorch, scikit-learn, pandas, SciPy, Plotly, XGBoost, LightGBM.
Math	Statistics, probability, stochastic processes, time series analysis, probabilistic ML, Bayesian analysis.